

Get-Rescue — Project Documentation & Status Report

Prepared 9 October 2026. Covers the Get-Rescue roadside-assistance dispatch application built on Zoho Creator.

Executive summary

Get-Rescue is a complete dispatch CRM for roadside assistance, covering both RSR (on-site repair) and TOW service types. It is built as twelve Zoho Creator widgets over a single `Create_Case` record per ticket, supported by twenty server-side Custom APIs.

The originally scoped application — agent intake wizard, vendor/technician/driver field apps, operations board, quoting, payments and closure — is **built and in client testing**. Since the original scope was agreed, the client has introduced several substantial additions: a per-vendor invite architecture, an agent-on-behalf-of-vendor data entry mode, a detailed locations/distances/timestamps specification, telephony integration, a public customer page, performance dashboards, and a vendor accounting and operations-approval layer. All of these have been built.

What remains is a small number of items awaiting client decisions (chiefly billing rules on cancelled cases), two Zoho configuration steps, one WhatsApp template text change that is blocking a completed code change from going live, and systematic end-to-end testing. These are itemised in sections 7 and 9.

Evidence base and its limits

This report is compiled from the project repository — the per-widget working logs (`README.md`), the field reference (`FIELDS.md`), the access matrix (`ACCESS.md`), the widget source code and the Custom API source — together with the client specifications currently available to the author.

Limitation, stated plainly: several client documents referenced in the project history were supplied at various points and are not all available for direct re-reading at the time of writing. These include the original Screen-by-Screen Flow Atlas, the Action 1–7 flow specifications, the LOCATIONS-DISTANCES-DATETIME specification of 7 September, the Ozonetel integration document, the vendor-accounting specification and the soft-test feedback lists. Where this report describes those requirements, it does so from the implementation record written at the time each was acted on — which is contemporaneous and detailed, but is a secondary source. Items that could not be verified against either a primary document or the live code are marked **Needs Verification** and should be confirmed before being presented as fact.

The client’s edited “Google Distances and Times” specification and covering email of 8 October, and the five-point multiple-vendor requirement list, were available in full and are reported directly.

1. Project Overview

Scope. A dispatch and case-management system for a roadside-assistance operator, covering the full life of a case from the customer's first call to vendor payment and closure, for two service types: **RSR** (a mechanic repairs the vehicle on the spot) and **TOW** (a vehicle is recovered and delivered to a drop location).

Architecture. Zoho Creator is the backend. Each ticket is one `Create_Case` record, created on first save and updated at every subsequent step. `Create_Case.Status` is a fixed sixteen-value picklist; every stage writes one of those exact values. The user interface is a set of self-contained HTML widgets embedded in Creator, talking to it through the widget SDK Data API, plus twenty Deluge Custom APIs for work that must run server-side (payments, invoicing, webhooks, telephony, cross-report reads).

Modules

Module	Audience	Purpose
<code>getRescueTicket</code>	Agent / back office	The core step wizard. Creates and drives a ticket through its full lifecycle: Create → Quote & Booking Fee → Capture Locations → Vendor Assignment → RSR or TOW path → Final Payment → Final Closure. Also hosts the dashboard, search, Ops Map, Accept/Reject audit, vendor accounting and operations approval.
<code>vendorTicket</code>	Vendor	A vendor's full lifecycle across both service types, with an Individual path (the vendor does the job) and a Fleet path (accept, then hand off to one of their own technicians).
<code>technicianTicket</code>	Technician	The RSR on-road flow: Accept/Reject, Reach, Work In Progress, Final Payment.
<code>driverTicket</code>	Driver	The TOW on-road flow: Accept/Reject, Reach, Loading, Reached Drop, Unloading & Handover, Final Payment.
<code>Ticket Kanban</code>	Operations	A board view over existing cases, one column per Status, reusing the same edit wizard.
<code>customerPage</code>	Customer (public, no login)	Booking-fee payment, remaining payment, location sharing and feedback. Routed through narrow Custom APIs rather than direct report access, to avoid a per-record data-exposure risk in Creator's public-page mechanism.
<code>vendorPerformance</code>	Vendor	Read-only aggregate statistics for the logged-in vendor. Never lists individual tickets.
<code>agentPerformance</code>	Agent	Read-only aggregate statistics for the logged-in agent, filterable by date range.
<code>callCenter / inboundCallPop</code>	Agent	Voice callbacks and inbound-call handling through Ozonetel, with a call log.

Module	Audience	Purpose
serviceRequest	Agent	Service-request intake. Needs Verification — present in the repository with its own working log; its current role relative to `getRescueTicket` is not restated here.
toggleGetRescue	—	The original standalone availability switch. Superseded 31 July by a toggle built into each field app; retained for reference only.
vendorPerformance, agentPerformance, callCenter	—	See above — all added early September.

2. Original Requirements

The original scope, as recorded at the time it was built, covered the following.

Area	What was required
Agent ticket wizard	A guided, step-by-step screen set that creates one case and carries it through its life. Steps: Create Service Ticket; Service Fee Quote & Booking Fee; Capture Locations; Vendor Assignment; then a service-type branch into either the RSR path (Service Acceptance, Reach Breakdown Location, Work In Progress) or the TOW path (Service Acceptance, Reach Breakdown, Loading, Reached Drop Location, Unloading & Handover); then Final Payment Collection; then Final Closure.
Status model	A fixed sixteen-value Status picklist driving both the case's own state and where the wizard resumes when a case is reopened. Statuses include SERVICE FEE QUOTE, LOCATION REQUEST, BOOKING FEE PENDING, LOCATION PENDING, READY FOR ASSIGNMENT, RSP IRA (INVITE RESPONSE AWAITED), RSP ON THE WAY, WORK IN PROGRESS, LOADING, TO DROP LOCATION, UNLOADING, REMAINING FEE DUE, RSP CLOSED, RSP CANCELLED, RSP REJECT, CX REJECT.
Quoting	A fee quote driven by a Rate Master, keyed on vehicle type, vehicle segment and the selected vehicle issue, producing a suggested rate and a rate breakup. Flat-rate and distance-formula rate rules. An agent-entered total service fee, a booking fee collected up front, and a remaining fee due at completion.
Locations	Capture of the breakdown location, and for TOW also a drop location, as coordinates on the ticket. A link sent to the customer to share their location. A round-trip distance calculated for quoting.
Vendor assignment	A vendor picker filtered by distance from the breakdown location (8 km for RSR, 20 km for TOW), sorted by priority then distance, showing each vendor's priority, distance, phone and online/offline state. A preferred-vendor override that shows every vendor.
Field applications	Separate applications for the vendor, technician and driver, each showing only their own assigned cases, with an online/offline availability toggle and the on-road actions for their service type, including photo capture at defined points.
Final Closure	A dedicated closure screen, separate from the wizard and reachable at any point, recording closure status and reason, refund details, customer feedback score and text, B2B GST invoice requirement and GST number, and a closure photo. It also surfaces every field from every stage as an individually editable row.

Area	What was required
Operations board	A Kanban view over existing cases, one column per Status, reusing the same edit wizard.
Distance calculation	Road distance and travel time between defined points, using the Google Maps Distance Matrix API.

3. Completed Requirements

Grouped by module. “Completed” here means built and present in the deployed code; section 9 separates this from independently tested and confirmed.

3.1 Agent ticket wizard

Requirement	What was built
The step wizard	All twelve stages are implemented as a single <code>STAGES</code> table driving one shared renderer. The path branches on <code>Service_Type</code> after Vendor Assignment. A stepper shows progress and locks steps ahead of unsaved work.
Status writing and resume routing	Every stage computes and writes one of the sixteen picklist values. Each stage declares the statuses it resumes from, so reopening a case lands on the right step. A guard (<code>statusStillEarly</code>) prevents any stage from recomputing a Status backwards once the case has genuinely moved past it.
Quoting	Rate Master lookup by vehicle type, segment and issue; flat and distance-formula rules; suggested rate; auto-generated rate breakup that re-seeds when the issue selection or quote distance changes but preserves a manual edit otherwise; remaining fee computed as total minus booking fee.
Locations	Coordinate capture by paste-and-split into separate latitude/longitude fields, for both breakdown and drop. A customer location-request link sent by WhatsApp. Round-trip distance calculated for quoting. <code>Location_Received_Time</code> stamped once, on first capture.
Vendor assignment	Distance- and priority-sorted vendor picker with the RSR/TOW radius split and the preferred-vendor override; each option shows priority, distance, phone and availability.
Final Closure	The dedicated closure screen with all ten closure fields, the full per-stage editable summary, collapsible groups, and a view-only lock once the case is closed.
Dashboard and search	Four status-grouped report lists, per-row paid/location flags, and search by phone, Case ID, vehicle registration number and client tracking ID.

3.2 Field applications

Requirement	What was built
Per-user case lists	Each app resolves the logged-in user’s own record and shows only their cases. The vendor app covers both

Requirement	What was built
	service types and both the Individual and Fleet paths, including hand-off to one of the vendor's own technicians.
Availability toggle	An online/offline switch built into each of the three field apps, writing to the user's own availability record. This replaced the original standalone toggle widget.
On-road flows	RSR: Accept/Reject, Reach, Work In Progress, Final Payment. TOW: Accept/Reject, Reach, Loading, Reached Drop, Unloading & Handover, Final Payment. Each step captures the photographs its stage requires, with enforced minimum counts.
GPS capture	Position recorded at Accept, Reach, Cancel, Picked-Up and Reached Drop, each written to its own field on the case.
New-case alerts	A background poll while the case list is on screen detects a newly arrived invite or a status change on a known case, and raises an audible alert. Each of the three apps uses a deliberately distinct sound.

3.3 Distances, ETA and timestamps

Nine measurements are produced, each captured once at a defined trigger and then held. Locations A–H from the client's specification are all captured and saved: drop location, vendor reach-drop position, breakdown location, vendor start position, vendor reach position, vendor cancel position, nearest vendor fleet location, and the static office coordinate.

Measurement	Trigger	Status
Start-to-breakdown distance and ETA (RSR and TOW)	Vendor taps Accept	Completed, on Google
Breakdown-to-drop distance and ETA (TOW)	Vendor taps Picked-Up	Completed, on Google
Rescue round-trip distance (Roundtrip_Distance)	Task Confirmation Time, as of 8 October	Completed
Vendor distance (Vendor_Distance, RSR)	Agent Closure save	Completed — one branch outstanding, see section 7
Vendor round-trip distance (Vendor_Round_Trip_Distance, TOW)	Agent Closure save	Completed — one branch outstanding, see section 7

Measurement	Trigger	Status
Cancel distance and cancel-to-breakdown distance	Vendor cancels in their app	Completed; cancel-to-breakdown uses straight-line by design
Travel mode	—	Completed — TOW routes as four-wheeler, RSR as two-wheeler

3.4 Payments and invoicing

Requirement	What was built
Booking fee collection	A payment link generated through Zoho Payments and sent to the customer on an agent click, with a public customer page to pay on. A webhook records the outcome, writes the transaction reference, the payment method and the receipt time, and advances the case status.
Remaining fee collection	The same mechanism at the Final Payment stage, against its own fields.
On-demand invoicing	Invoice creation in Zoho Books for on-demand (walk-in) cases, with the GST split applied, payment recorded against the invoice, and the invoice PDF captured back onto the case and openable from the ticket.
Vendor payment	Vendor fee calculation from the vendor's own rate card — a KM slab formula applied to the actual distance, plus night-rate handling and extras — producing gross fee, TDS and net payable. TDS resolved from the vendor's own Zoho Books record with a rate cross-check. Payment recorded back to Books.
Operations approval	A review stage where an operations manager can adjust the TOW round-trip distance and extras, see the fee recalculate live, and approve — individually or in bulk — with the computed fee written back and the settlement status advanced.

3.5 Messaging, telephony and the customer page

Requirement	What was built
WhatsApp messaging	Templated messages through WhatsPortal at defined points in the flow: the opener pair, booking-fee confirmation, rescuer details and ETA, an extra-charges advisory, status milestones and a Google feedback request. Customer-facing sends are deliberately placed on an agent click rather than firing automatically on

Requirement	What was built
	reaching a step.
WhatsApp inbound webhook	A webhook endpoint that accepts three different payload shapes, matches an inbound message to a case on the last ten digits of the sender, and writes an inbound log.
Ozonetel telephony	Click-to-call from a ticket, inbound screen-pop handing the call identifier to the ticket, automatic and manual linking of a call to a case, and a per-ticket call history that matches on the case link, the call identifier or the phone number.
Customer page	A public, no-login page for booking payment, remaining payment, location sharing and feedback, routed entirely through narrow Custom APIs.

4. New Requirements and Client Changes

Everything below was introduced or changed after the original scope was agreed. The distinction between a genuinely new requirement and a change to existing behaviour is noted in each entry.

4.1 New requirements

Date	What the client introduced	Type
3 August	Per-vendor invite architecture. One <i>Invites</i> record per invited vendor per case, created when the Assignment step is saved, carrying that vendor's own response. Replaced reading assignment state off the shared case record.	New — a new form and a new data flow
7 September	LOCATIONS-DISTANCES-DATETIME specification. A formal definition of every location to capture, every distance and ETA to calculate, each one's trigger point, and a set of named timestamps. The primary document is not available for re-reading; this entry is from the implementation record.	New — substantially extended an existing area
Early September	Customer page, vendor performance dashboard, agent performance dashboard. Three new user-facing surfaces.	New
September	Ozonetel telephony integration. Click-to-call, inbound screen-pop, call logging and call-to-case linking.	New
September	Vendor accounting and operations approval. Vendor fee calculation, TDS, settlement status and an approval stage.	New
October	WhatsApp inbound webhook. Receiving customer replies rather than only sending.	New
October	Invoice PDF display. The Zoho Books invoice PDF itself shown from the ticket, rather than a link to a Books page.	New

4.2 Changes to requirements already built

Date	Originally	Changed to
7 August	Eight vendor/technician/driver stages were entirely	Every field editable at every step , so an agent can enter a case

Date	Originally	Changed to
	view-only for the agent.	on a vendor's behalf. Implemented through an <code>Acceptance_Mode</code> setting that switches each field between agent-editable and locked, rather than a blanket unlock.
3 October	A rejected case stayed on the vendor's list for 24 hours so they could see what they had turned down.	It must disappear immediately. The window was removed rather than shortened.
3 October	A reject wrote <code>RSP REJECT</code> onto the shared case record.	A reject goes on that vendor's own invite row only. The shared case is marked rejected only when no other invited vendor is still live — so one vendor's reject no longer blocks the others.
8 September, then 7 October	Google Distance Matrix was used for every distance calculation in every app.	First restricted to closure calculations only (8 September), then re-enabled for four named calculations and no others (7 October). Implemented as an explicit opt-in per call site, defaulting to straight-line, so no unlisted calculation can silently return to billed API usage.
7 October	The booking-fee payment method defaulted to Cash and a receipt photo was mandatory whenever a method was set.	A zero booking fee requires no receipt and leaves the method blank. Extended 8 October: a fee already paid through the gateway no longer pre-selects Cash either.
8 October	<code>Roundtrip_Distance</code> was calculated at Agent Closure.	Calculated at Task Confirmation Time. Implemented as an additional, earlier write; the closure calculation is retained as a fallback and can never overwrite the earlier value.
8 October	The customer's ETA was sent as a number of hours.	Sent as a clock time , calculated ETA plus ten minutes for vendor-app cases, or acceptance time plus the agent's ETA override for agent-on-behalf cases.

4.3 The multiple-vendor requirement list

Five points were raised together. All five are implemented; points 1 and 3 were blocked until 9 October by a single defect in the invite-response write path, described in section 7.

#	Requirement	Status
1	When the first vendor accepts, the case disappears from every other invited vendor's app and they are told they missed it.	Completed
2	A vendor who does not respond within two minutes times out, the case disappears, and they are told it was missed.	Completed
3	Multiple-vendor assignment status must be visible on the ticket.	Completed — the Vendor Invite Status panel on the Assignment step, plus a dedicated Accept/Reject audit screen
4	A rejected case must disappear from the vendor's app.	Completed
5	One vendor's reject must not mark the case rejected for the others.	Completed

5. Original vs. Changed vs. Implemented

Module / Feature	Original Requirement	Client's New Requirement or Change	What We Implemented	Status	Remarks / Pending
Vendor assignment	A vendor picker filtered by distance and priority; assignment written to the case.	Per-vendor invites; first-accept wins; two-minute timeout; per-vendor reject; visible status.	An <code>Invites</code> record per vendor per case; invite status panel on the Assignment step; a separate Accept/Reject audit screen with response time and reason.	Completed	Depends on <code>Responded_At</code> being on the Invites report layout — done 9 October.
Agent-on-behalf entry	Vendor stages were view-only for the agent.	Every field editable so an agent can run the case for the vendor.	<code>Acceptance_Mode</code> switches each field between agent-editable and locked; distance calculations branch on the same setting.	Completed	—
Distances and ETA	Google Distance Matrix for distance and travel time.	A formal nine-measurement specification with named trigger points; then Google restricted to four named calculations; then revised branch rules and trigger points.	All nine measurements built with their trigger points; Google opt-in per call site; branch logic per the revised specification.	Completed	Two branch rules await a client billing decision — section 7.
Customer ETA message	ETA shown to the customer.	Clock time, +10 minutes on vendor-app cases; acceptance time + override on agent-on-behalf cases.	Both rules built, 24-hour format, with a switch to revert.	Blocked	The approved WhatsApp template has the word “Hrs” in its body; it must be edited and re-approved by Meta.

Module / Feature	Original Requirement	Client's New Requirement or Change	What We Implemented	Status	Remarks / Pending
Booking fee	A booking fee collected up front, by link or in person, with a receipt photo.	Zero fee requires no receipt and no method; a gateway-paid fee must not demand a cash receipt.	The receipt requirement and the method default are both conditional on the fee and on whether it was already paid; on-screen text states the rule in force.	Completed	Add "Payment Gateway" as an option on the `Payment_Method` field in Zoho.
Invoicing	Not in the original scope.	On-demand invoicing in Zoho Books; then the invoice PDF shown from the ticket.	Invoice creation with GST split, payment recorded, PDF captured and openable from the ticket.	Completed	—
Vendor payment	Not in the original scope.	Vendor fee from a rate card, TDS, settlement status, operations approval.	KM slab formula, night rate, extras, gross/TDS/net, Books payment, approval stage with bulk approval.	Completed	Ops_Approval_Comments must be changed from Number to Multi Line in Zoho.
Telephony	Not in the original scope.	Ozonetel click-to-call, screen-pop, call logging.	All three, plus call-to-case linking with a warning when the call's number does not match the ticket.	Completed	—
WhatsApp	Customer notifications.	Inbound replies received as well as sent; booking-fee message after the customer taps Continue.	Templated sends at defined points; an inbound webhook writing a log.	Partially completed	The send after Continue is recorded but deliberately left on an agent click. The client expects it automatic — needs confirmation.
Closure	A dedicated closure	Closure-time distance	All three, plus a per-stage	Completed	Ten closure field

Module / Feature	Original Requirement	Client's New Requirement or Change	What We Implemented	Status	Remarks / Pending
	screen with refund, feedback and GST fields.	calculations; validation rules; a view-only lock once closed.	editable summary of every field in the case.		names were originally guesses; several have since been confirmed in use. Needs Verification for the remainder.

6. Additional Features and Enhancements

Built beyond the stated requirements, each in response to a real failure or risk observed during the project.

Enhancement	Purpose and benefit
Record locking	A case being edited is locked to that agent, with a heartbeat and a stale-lock timeout. Added after the client reported two agents editing the same new ticket simultaneously. Prevents one agent's work silently overwriting another's.
Offline photo queue	Photographs taken without a connection are queued and flushed when connectivity returns, with the submit path waiting on the flush. Added after photographs were reported lost when a vendor came back online.
Two-tap confirmation	Destructive actions in the field apps use an arm-then-confirm button instead of a browser confirmation dialog, because such dialogs silently return "no" inside mobile web views — which made photo removal appear broken on phones.
On-screen diagnostics	Where a failure would otherwise be invisible — an unresolvable login, a case with no linked invite, an invite with no readable ticket reference — the cause is rendered into the screen in plain language rather than only written to a console that does not exist on a phone.
Write verification	Several writes are read back and confirmed rather than trusted. Creator can accept an update that silently stores nothing, most often when a field exists on a form but not on the report layout. This pattern has caught real failures on the call link, the invoice PDF and the payment webhook.
Invite de-duplication	Re-saving the Assignment step no longer creates duplicate invite records.
Call-link mismatch warning	Linking a call whose number does not match the ticket is still possible — a customer may call from another phone — but is now marked on the button and spelled out in the confirmation, after a mismatched call was linked unnoticed.
Vendor base fallback	Where a vendor's measured distance is missing, the accounting screen falls back to the recorded breakdown distance so a case with a good measurement is never priced at zero kilometres.
Audible new-case alerts	Each field app raises a distinct sound when a new case arrives, generated in the browser with no audio file to host.
Search breadth	Cases are findable by phone, Case ID, vehicle registration number and client tracking ID, including closed cases

Enhancement	Purpose and benefit
	that are hidden from the dashboard.

7. Partially Completed and Pending Items

7.1 Awaiting a client decision

Item	What is needed
`Vendor_Distance` branch 3	The specification says that where neither a reach position nor a cancel position exists, the distance is the vendor's base to the breakdown location, for all repair cases with no status condition. Taken literally this pays a kilometre fee on cases cancelled before the vendor left base. Not implemented — this is a billing decision, not a technical one.
Vendor round-trip branch 4	The specification covers “cancelled billable or cancelled not-billable”. Only the billable case is handled. Extending it means paying a kilometre fee on non-billable cancellations — confirmation needed.
Vendor round-trip branch ordering	One branch is placed last rather than second so that it can only add a figure to cases that produce none, never change one already being paid. The literal specification order would alter existing cancelled-billable cases. One line to change if the client wants the literal order.
Field naming	Five fields are named differently in the specification than in the system. A mapping table is proposed rather than renaming fields that hold production data and are referenced by reports and exports.
Breakdown-to-drop origin	The specification measures from the vendor's stored reach position; the system measures from their live position at Picked-Up. The two are metres apart in practice.
`Office_To_Mechanic` / `Breakdown_To_Office`	The specification lists these as typed by the agent; the field apps calculate and write them. Which behaviour is correct?
WhatsApp booking-fee send after Continue	Currently on an agent click by standing instruction; the client expects it automatic.
Two contradictions between the client's email and their own attachment	The email objects to <code>Vendor_Distance</code> and asks for vendor click locations in the rescue round trip; the attached specification keeps the first and specifies agent-entry locations for the second. Confirmation needed on which governs.

7.2 Blocked on an external action

Item	Blocker
Customer ETA as a clock time	Built in all four applications. The approved WhatsApp template <code>share_rescue_service_details_and_eta</code> has the literal word “Hrs” in its body after the ETA variable, so a clock time would read “15:35 Hrs”. The template text must be edited in WhatsPortal and re-approved by Meta. A single switch reverts to the previous format in the meantime.

7.3 Zoho configuration outstanding

Change	Consequence if not done
Add Payment Gateway as an option on <code>Create_Case.Payment_Method</code>	The payment webhook’s write of the method is refused and it falls back to writing everything else, leaving the method blank.
<code>Ops_Approval_Comments</code> : Number → Multi Line	Approval notes are rejected; the approval itself still saves.
<code>Locked_By</code> / <code>Locked_At</code> on the <code>Agent_Ticket_Report</code> layout	Record locking cannot read back the lock it writes.
<code>Vendor Email</code> and <code>Base_Location_1_Lat</code> / <code>_Lon</code> on <code>Vendors_Report</code>	Vendor identity and the nearest-base calculation fail for affected vendors.

7.4 Known data condition

Of 152 invite records, 133 carry no usable reference to their ticket. Those were saved before the reference write was corrected; 19 resolve correctly and every new invite does. The affected rows show a blank Ticket column on the Accept/Reject screen. No remediation has been agreed — the historic rows can be left as they are or backfilled.

7.5 Resolved during this review

A defect in the invite-response write path, found and fixed 9 October. When more than one vendor was invited to a case, the code that identifies which invite row belongs to the responding vendor compared against a display name that Creator does not return — it returns a bare record identifier. The comparison was therefore always against an empty value, no row matched, and the response was never written. The case still moved correctly for the vendor; what failed was the office’s record of it. This is why an accepted case continued to show as pending, and why points 1 and 3 of the multiple-vendor list appeared not to work. Matching is now done on the record identifier, and a failure to record a response is reported on the vendor’s screen rather than only to a console.

8. Integrations and Technical Changes

8.1 External systems

System	Used for
Zoho Creator	The entire backend: forms, reports, records and the Data API the widgets run on.
Zoho Payments	Booking-fee and remaining-fee payment links, with a webhook recording the outcome.
Zoho Books	On-demand customer invoicing with GST, invoice PDF retrieval, vendor contacts, TDS resolution and vendor payment recording.
WhatsPortal (WhatsApp Business)	All templated customer messaging, and an inbound webhook for customer replies.
Ozonetel	Outbound click-to-call, callback status, inbound caller information and screen-pop.
Google Maps Distance Matrix	Road distance and travel time, restricted to four named calculations, with a straight-line fallback whenever a route cannot be obtained.

8.2 Server-side Custom APIs

Twenty Deluge Custom APIs carry the work that cannot safely run in a widget — anything needing a server-side credential, a cross-report read a user's profile cannot perform, or an endpoint an external system calls.

`createTicket`, `searchTickets`, `getVendorTickets`, `getTaskRejections`, `getVehicleNames`, `getInboundCallerInfo`, `recordRingingCall`, `getVoiceCallbackDetails`, `getCallbackStatus`, `getCallbackStatusJson`, `cleanCallLogUcids`, `testCallLogReport`, `generateZohoPaymentLink`, `paymentWebhook`, `onDemandInvoicing`, `fetchInvoicePdf`, `recordVendorPayment`, `sendWhatsAppMessage`, `whatsappWebhook`, `checkOperationsManagerPermission`.

8.3 Significant technical decisions

Decision	Why
Coordinates stored as text, not decimal	Creator rejected high-precision coordinates as exceeding a decimal field's digit limit. Four coordinate fields were converted to single-line text, which has no such limit.
Record identifiers sent as strings	A nineteen-digit Creator record identifier exceeds JavaScript's safe integer range. Sending it as a

Decision	Why
	number has corrupted a value in this project before.
Server-side filters avoided on lookup fields	Creator rejects a filter expression against a lookup field with an error that reads identically to “no matching records”. Every such read fetches and filters in the widget instead.
The customer page uses narrow Custom APIs	Creator’s public-page mechanism would otherwise expose more per-record data than the page needs.
Google usage is opt-in per call site	The helper serves several calculations and only four are approved for billed API usage. Defaulting to straight-line means a calculation added later cannot silently incur cost.
Customer-facing sends are on an agent click	A standing instruction: payment and WhatsApp messages go out when an agent chooses, not automatically on reaching a step.

9. Testing and Validation

This section is deliberately conservative. The project has no automated test suite. Validation has taken two forms: a syntax and structural check run against every widget before each release, and live testing by the client, whose reports have driven a large share of the changes recorded in this document.

9.1 Confirmed by live use

The following are confirmed working because a real failure in each was reported, fixed, and the fix confirmed in use: ticket creation and the full wizard path for both service types; status writing and resume routing; the dashboard and search; vendor assignment and the distance-sorted picker; the field apps' case lists, availability toggle and on-road steps; photo capture and upload including the offline queue; booking-fee payment links and the payment webhook; on-demand invoicing and invoice PDF retrieval; WhatsApp template sends; Ozonetel click-to-call and call logging; record locking; and the two-minute invite timeout.

9.2 Implemented but not yet confirmed by a completed test

Area	What needs testing
Multiple-vendor accept and reject	The 9 October fix to the invite-response write path has not yet been exercised by the client. Test with two vendors invited to one case: one accepts, the case should leave the other vendor's app, the Accept/Reject screen should show Accepted with a response time, and the Assignment step should name the accepting vendor.
`Roundtrip_Distance` at Task Confirmation	Confirm the value is present before closure on a repair case, and that a tow case confirmed without a drop location is still filled at closure.
`Vendor_Distance` cancel branch	A repair case cancelled by the vendor in their app should carry the cancel distance after closure.
Vendor round-trip branch 2	A tow where the vendor reached but never recorded a drop position should produce a base-to-reach-to-base figure.
Customer ETA as a clock time	Cannot be tested until the WhatsApp template text is changed.
Booking-fee rules	A gateway-paid case should show no payment method and demand no receipt; a zero-fee case should state that no receipt is required.
Call-link mismatch warning	Opening a case with no matching call should show a marked button and a confirmation naming both numbers.

9.3 Not independently verified

A number of field link names, option lists and report names in less-exercised areas were originally pattern-matched rather than confirmed against a live record, and some have never been exercised since. The project's field reference marks each of these. Any of them may fail silently, because Creator accepts an unrecognised field in a payload without error. These should be confirmed in Zoho Studio before the corresponding features are relied on commercially.

10. Final Project Summary

****	Summary
Original scope	An agent ticket wizard across twelve stages for two service types, a sixteen-status model, rate-master quoting, location capture, a distance- and priority-ranked vendor picker, three field applications, a Kanban operations board, and a dedicated closure screen. Built in full.
Completed work	The original scope, plus every major addition the client has introduced since: per-vendor invites with accept, reject and timeout handling; agent-on-behalf data entry; the full nine-measurement distance and ETA specification; payments, on-demand invoicing and invoice PDF display; vendor fee calculation, TDS and operations approval; WhatsApp messaging in both directions; Ozonetel telephony; a public customer page; and two performance dashboards.
Additional features	Record locking, an offline photo queue, mobile-safe confirmations, on-screen diagnostics, write verification, invite de-duplication, call-link mismatch warnings, a vendor-distance fallback, audible new-case alerts and broad search. Each was built in response to an observed failure or risk rather than a stated requirement.
Client-requested changes	Eight significant changes to already-built behaviour, the largest being the shift from view-only vendor stages to fully agent-editable entry, the move from a shared reject to per-vendor invites, two successive redefinitions of where the Google API may be used, and the revised distance branch rules and trigger points of 8 October.
Remaining work	Eight client decisions (section 7.1), one WhatsApp template change blocking a completed feature (7.2), four Zoho configuration changes (7.3), and the testing listed in 9.2. No remaining item represents a substantial new build; the largest is a few lines of code once a billing rule is confirmed.

Status at a glance

Area	Status
Agent ticket wizard	Completed
Field applications (vendor, technician, driver)	Completed
Multiple-vendor assignment, accept, reject, timeout	Completed — awaiting client test of the 9 October fix
Distances, ETA and timestamps	Completed — two branch rules await a billing decision

Area	Status
Customer ETA message format	Blocked on a WhatsApp template change
Payments, invoicing and vendor accounting	Completed
WhatsApp messaging	Completed — one send deliberately manual, pending client confirmation
Telephony	Completed
Customer page and performance dashboards	Completed
Operations board	Completed — a view-only lock on reopening a closed case is not yet built